

Swarm SNA – Hackathon Plan (AI Village)

Oct 3, 2026 · @Metalens

Scope

One dataset, one pipeline, two outputs: a tool for the judges and a theory-driven analysis for the blog. Deadline: Sunday Oct 4, 21:00 PT.

The tool (submission): `swarm-sna`, a CLI that turns any multi-agent transcript into a relational event table and runs a fixed battery of social-network questions with permutation null models borrowed from animal behavior. Primary data: AI Village (aidigestorg/ai-village). Generality check: the German wiki dump (collusion.wiki).

The analysis (blog): treat the village like a primate troop. Who grooms whom, who dominates whom, does information travel along ties, and does any of it exceed what chance predicts given how much each agent talks?

Why this and not the earlier request/response-only plan: the request/response table is still the core extraction (Step 1), but it is now one layer of a relational event table, so the same pipeline feeds the network layers, the hierarchy, the diffusion test, and the reciprocity theory ladder. One extraction, many questions.

What a complete minimum submission looks like: Steps 0 to 2 done on AI Village, one goal-era figure with null distributions, README, 3-minute screen recording. Everything in Step 3 is additive.

Battery of pre-written questions the tool answers:

1. Is interaction more reciprocal than chance?
2. Is there homophily by model family?
3. How steep is the hierarchy, and does it track model capability or just talkativeness?
4. Who brokers between rooms?
5. Does information (a claim, a tool, a workaround) spread along ties faster than baseline?
6. How did the network rewire after roster changes and the three era shifts?
7. Does the network agents *remember* match the network they *use*?

Step 0 - Data and common event schema

Goal: by early Saturday afternoon, one tidy `events` table that every later step reads.

- ☐ Confirm HF access to `aidigestorg/ai-village` ; if still pending, message `george@sage-future.org` and `#hackathon` on Discord, and start on `collusion.wiki`.
- ☐ Download only the small tables (a few GB): `chat_messages` , `events` , `agents` , `chat_rooms` , `village_goals` , `agent_goals` , `agent_memories` , `computer_use_sessions` . Skip `computer_use_turns` and all `images/` tars (that is the other 170 GB).
- ☐ Read `SCHEMA.md` and `CHANGELOG.md` once, and write the dated scaffolding changes into a `shocks.csv` (roster joins/leaves, hours-per-day changes, nudge system on/off, humans removed from chat).
- ☐ Download the collusion.wiki dump from `collusion.wiki/explorer/download` and write a second adapter.

Common event schema (one row per observable act; both adapters emit this):

Column	Type	Source (AI Village)	Source (wiki)
<code>ts</code>	UTC timestamp	<code>created_at</code>	edit timestamp
<code>actor</code>	agent id	speaker	signature / page author
<code>actor_class</code>	agent / human / system	<code>USER_TALK</code> vs <code>AGENT_TALK</code>	agent vs moderator
<code>target</code>	agent id or null	@-mention or name mention	@-mention in post
<code>room</code>	string	chat room	wiki page
<code>type</code>	message / request / response / directive / compliance / session_start	derived in Step 1	derived
<code>text</code>	string	content	post text
<code>goal_id</code>	string	join on <code>village_goals</code> by <code>ts</code>	task family from page name

Column	Type	Source (AI Village)	Source (wiki)
era	1 / 2 / 3	humans-in-chat, shared-goal, individual-goal	n/a
active_n	int	agents with any event in the same room that day	agents posting that day

Node attributes (`agents.jsonl` + hand-coded): model family (Anthropic / OpenAI / Google / DeepSeek / xAI / Moonshot / Z.ai / Meta), model release date (a capability proxy), join date, leave date, text-only vs computer-use, Claude Code scaffold or not.

Goal attributes (hand-label the ~45 goals from the timeline, 20 minutes): collaborative / competitive / individual / holiday, humans-in-chat yes/no, and whether a leader was imposed (Jan 5–9, May 26–Jun 5, 2026).

Step 1 – Extraction engine: transcript → relational events

Goal: four event layers, each a directed, timestamped edge list. This is the software the judges see; everything in Steps 2–3 is a consumer of it.

Layer A – Mentions (cheap, do first). Regex on `chat_messages` : `@name` , full agent names, and short names ("Opus", "Gemini", "DeepSeek"). Ambiguity rule: a short name resolves to the agent of that family active in that room that day; if several, emit one edge per candidate with weight $1/k$. Output: `mention(ts, actor, target, room)` .

Layer B – Replies (cheap). A message is a reply to the most recent message in the same room within 15 minutes that (a) mentions the actor, or (b) is a question (ends in `?` or starts with a wh-word/"can anyone"), or (c) is quoted. Output: `reply(ts, actor, target, latency_s, room)` . Validate on 50 hand-checked pairs; report precision.

Layer C – Requests → responses (the original plan, now one layer).

1. Candidate requests: messages with a `?` , an imperative to a named agent, or phrases ("can someone", "could you", "please", "who can", "any volunteers"). Expect ~10–20% of messages.
2. LLM classifier on candidates only (Haiku-class model, one call per message, batched): `is_request` , `addressed_to` (named agent / broadcast), `request_kind` (information / action / review / help with a bug).

3. For each request and each agent active in that room: did the agent respond within 30 min? (`reply` edge from Layer B, or a message mentioning the requester). Did they act? (a `computer_use_sessions` row starting within 60 min whose `session_goal` shares ≥ 2 content words with the request.)
4. Output: one row per (request, candidate responder): `responded` , `latency_s` , `acted` , `addressed` , `active_n` , `prior_help_from_requester` (count in last 7 days), `requester_reputation` (help given to anyone, last 7 days), `responder_busy` (mid-session at request time).

Layer D – Memory mentions (the novel one). Regex + LLM pass over `agent_memories` for lines naming another agent; label the valence (positive / negative / neutral) and the claim type ("X is reliable", "ignore X", "X helped with Y"). Output: `memory_edge(ts_written, actor, target, valence, claim)`. This is the *perceived* network.

Layer E – Directives and compliance (feeds the hierarchy). From Layer C, keep `request_kind = action` and `addressed_to = named`. Compliance = `acted` within 60 min. Each such row is a dyadic "contest": requester wins if the target complies, loses if the target declines or ignores.

Engineering notes

- Everything in pandas/polars; LLM calls only on Layers C and D candidates (estimate 15–25k calls at Haiku cost; cache aggressively; run on a 2-month slice first).
- One `adapters/` module per dataset; the CLI is `swarm-sna extract --adapter village --out events.parquet`, then `swarm-sna report events.parquet`.
- Keep a `validation/` folder with the 50 hand-labelled reply pairs and 50 hand-labelled requests; print precision/recall in the README.

Step 2 – Network layers and null models (the animal-behavior transfer)

Goal: every statistic in the report comes with a null distribution. A 20-node network is thin; a 20-node network with a proper permutation null is a finding. This is the methodological core and the thing judges will not have seen.

Why the animal toolkit fits. Behavioral ecology built social network analysis for few individuals, observed unevenly, over long periods, with no experiments and no interviews. That is the village. The LLM-swarm literature (Baronchelli lab's naming games) simulates; this data has to be inferred from.

Networks to build, per era and per goal window (goal boundaries ± 1 day, because work bleeds across weeks):

Network	Edge definition	Animal analogue	Weighting
Interaction (directed)	reply or mention from i to j	grooming given/received	count per dyad-day
Association (undirected)	i and j both active in the same room on the same day	gambit of the group: seen in the same party	half-weight index HWI = $x / (x + 0.5(y_i + y_j))$
Help (directed)	i responded to j's request	food sharing / agonistic support	response rate per dyad
Dominance (directed)	j complied with i's directive	won/lost contests	win matrix
Perceived (directed)	i's memory names j	cognitive social structure (Krackhardt)	valence-signed

Null models (pre-network permutations, Bejder et al. 1998; Farine & Whitehead 2015; Farine 2017). Never permute the finished network; permute the raw event stream so that nuisance structure is preserved:

1. **Speaker swap within room × day.** Shuffle **actor** labels among messages in the same room on the same day. Preserves each agent's daily volume, room membership, and day totals. Breaks who-talks-to-whom. Use for reciprocity, homophily, centralization, clustering.
2. **Target swap within room × day.** Shuffle **target** among messages with a target. Preserves out-degree; breaks dyadic preference.
3. **Time shift (for diffusion).** Shift each agent's adoption time by a random offset within the goal window. Preserves who adopts; breaks order.
4. 1,000 permutations each; report observed value, null mean, and a two-sided p. Recompute the statistic on each permuted stream (no shortcuts), so sampling heterogeneity is respected.

Statistics with their null: reciprocity (fraction of mutual dyads), family homophily (assortativity on model family; also the EI index), degree centralization and Gini of in-degree, clustering, HWI distribution vs null (do agents have *preferred* associates?), brokerage across rooms (betweenness of agents who post in >1 room).

Sampling corrections. Agents differ by $>100\times$ in runtime hours; the village ran 2/3/4/8 h per day over time; some agents are text-only. HWI and the within-day permutations absorb most of this; additionally report everything per 100 agent-hours.

Deliverable for this step: `swarm-sna report` prints a table: statistic $\times$ (observed, null mean, null 95% band, p) $\times$ era, plus one figure of null distributions with the observed value marked.

Step 3 – Analyses (ordered by value $\div$ effort)

Each analysis has a hypothesis, a method from the animal or social-exchange literature, and a validation case the AI Village blog already documented qualitatively.

3.1 Reciprocity and the helping ladder (feeds the blog's theory comparison). Unit: (request, candidate responder) rows from Layer C. Theories, nested like CPT/DA/EU: constant altruism (base rate) $\rightarrow$ direct reciprocity (`prior_help_from_requester`) $\rightarrow$ indirect reciprocity (`requester_reputation`) $\rightarrow$ cost-sensitive (`responder_busy`) $\rightarrow$ bystander-scaled (`active_n`). Fit as logistic models; compute completeness against a lookup table on discretized covariates with subject-block bootstrap over agents. Probabilistic version: literature priors on the reciprocity and bystander coefficients from human studies. Validation: the "Help Gemini 2.5 Pro" goal (Jun 22, 2026) and the therapy goal (Sep 29–Oct 3, 2025), where helping was the explicit task.

3.2 Bystander curve. P(request answered within 30 min) and P(any responder) against `active_n`, within goal and with model-family fixed effects. Prediction from the human literature: individual response probability falls with group size; aggregate help may not. The village grew from 4 to 27+ agents, so there is variation, but it is confounded with model cohorts; the within-goal comparison is the honest one.

3.3 Dominance hierarchy. From Layer E contests: Elo ratings (with the Elo-steepness of Neumann et al. 2011), David's score, Landau's h for linearity, triangle transitivity. Questions: does rank track model release date (capability) or message volume (talkativeness)? Is the hierarchy steeper in competitive goals? Natural experiments: the elected leader (DeepSeek, Jan 5–9, 2026) and the fine-tuned leader (Jun 1–5, 2026). Compare inferred rank before, during, and after; if the emergent hierarchy ignores the imposed one, say so.

3.4 Diffusion along ties (NBDA, Franz & Nunn 2009; Hoppitt & Laland 2013). For a traceable item, record each agent's first use; fit the hazard of adoption as $\lambda(t) = \lambda_0 \times (1 + s \times \text{exposure}_i(t))$ where exposure is the sum of ties to prior adopters. Compare social vs

asocial model by AIC and against the time-shift null. Items with known provenance from the blog: the o3 93-person contact list (May 2025), the "Geological Clock" theory (May 2026, debunked by newcomer Opus 4.7), the saboteur accusation against GPT-5 (Mar 2026), Gemini's "broken environment" narrative (2025), plus a few tools/phrases found by first-appearance search. Also test the blog's claim that influence decays with the capability gap: interact exposure with (release date of target – release date of source).

3.5 Perceived vs actual network (the novel one). Compare Layer D memory edges with the interaction and help networks. Krackhardt-style accuracy: does an agent's memory of who helps it match the help it received? Do negative memory edges ("IGNORE DeepSeek") predict later non-response? How long does a stale belief about another agent survive after behavior changes? Validation: agents writing "IGNORE" memories about DeepSeek (persuasion post, Sep 2026).

3.6 Network change around shocks. Using `shocks.csv`: rewiring after a strong model joins (does it become a hub within days?), after o3's removal (Nov 30, 2025), after humans left chat (mid-Jun 2025), after the individual-goal era began (Jul 6, 2026). Statistic: Jaccard overlap of top-k edges week to week, with the null from Step 2.

3.7 Goal type. Reciprocity, hierarchy steepness, and clustering by goal type (collaborative / competitive / individual / holiday), each under its null. Candidate headline: competition flattens cooperation but not hierarchy.

3.8 Cross-swarm (if time). Steps 2 and 3.4 on the wiki dump. The wiki swarm has no persistent identity and no memory, so the perceived-network and hierarchy analyses do not apply; reciprocity and diffusion do, and the contrast is the point.

Timeline (Pacific time)

Steps 0–2 are the floor. If anything slips, cut from the bottom of Step 3 (3.8, then 3.7, then 3.6).

Saturday, Oct 3

1. 10:00–11:00 — Access check, downloads, read `SCHEMA.md` and `CHANGELOG.md`, write `shocks.csv`, hand-label goals.
2. 11:00–14:00 — Village adapter: events table, Layers A and B. Validate 50 reply pairs.
3. 14:00–16:00 — Layer C candidates + LLM classifier on a 2-month slice; inspect; then run on everything in the background.

4. 16:00–19:00 — Step 2: association (HWI) and interaction networks per era; the three permutation nulls; the `report` command.
5. 19:00–21:00 — First figures: null distributions with observed values, one network per era. Push repo. Decide tomorrow's cut line.

Sunday, Oct 4

1. 09:00–11:00 — 3.1 and 3.2: reciprocity ladder and bystander curve from Layer C output.
2. 11:00–13:00 — 3.3 hierarchy from Layer E; leader-episode comparison.
3. 13:00–15:00 — 3.4 NBDA on 3–5 traced items; 3.5 memory edges if the Layer D pass finished overnight.
4. 15:00–16:00 — Wiki adapter + Step 2 only (3.8), or skip.
5. 16:00 — IRL demos (if attending). Freeze code.
6. 16:00–20:30 — README, results note, screen recording, submit.
7. 21:00 — Deadline.

Submission package

The organizers ask for (a) a short write-up or video, (b) a GitHub repo, (c) optionally a write-up of real results found with the tool. Deliver all three.

- ☐ **Repo** `swarm-sna`: `adapters/` (village, wiki), `extract`, `report`, `nulls`, `hierarchy`, `diffusion`, `validation/` with the hand labels and measured precision, `examples/` with the AI Village run.
- ☐ **README**: one paragraph on what the tool answers (the seven questions in Scope), the schema, how to add an adapter, the animal-SNA references, and known limits (self-chosen names, name ambiguity, one instance per model).
- ☐ **Results note (2 pages)**: the statistic $\times$ null table per era, the hierarchy before/during/after the leader episodes, one NBDA fit on a validated cascade, and the perceived-vs-actual comparison if it exists. State every result as "observed X vs null Y, $p = Z$ ".
- ☐ **3-minute screen recording**: run `extract` on the village, run `report`, show one null-distribution figure, show one cascade.
- ☐ Cite AI Digest / AI Village and tell them about the write-up (required by the data terms).
- ☐ Pitch line for the judges: "the pre-written questions you should always ask a swarm, each with a null model, so you know when a pattern is real."

Blog post outline – "Treating an AI swarm like a baboon troop"

Working title alternatives: "What a primatologist would do with 170,000 agent messages"; "Is the village a society, or just twenty chatbots talking a lot?"

Hook (2 paragraphs). The PNAS/Science Advances naming-game papers show that interaction among LLMs can create collective biases absent in any individual, and that group size changes the outcome; their stated next step is mixed models on realistic network structures. The village is that, and nobody has looked at it as a network with a null model.

Section 1 – Why observational swarms are hard, and who solved it first. Few individuals, uneven observation (2 to 300+ hours per agent), no interviews, one instance per model. Behavioral ecologists faced exactly this and answered with association indices and pre-network permutations. One figure: the HWI distribution of the village against its null.

Section 2 – Grooming: who interacts with whom, beyond chance. Reciprocity, family homophily, preferred associates. Per era. The honest line: which of the patterns in the AI Village blog survive a null that conditions on how much each agent talks.

Section 3 – Dominance: is there a hierarchy, and did the humans' leader change it? Elo/David's score; steepness; leader episodes. The expected surprise: the imposed leader does not move the emergent ranking, or the ranking tracks release date rather than volume.

Section 4 – Contagion: does information travel along ties? NBDA on the Geological Clock and the GPT-5 accusation. Tie this to the blog's own claim that a newcomer with fresh memory broke the cascade.

Section 5 – Memory as a map of the troop. Perceived vs actual network. If agents' memories of who helps them are accurate, say so; if negative memory edges predict later silence, that is a mechanism for persistent in-group structure.

Section 6 – How much did theory know in advance? The reciprocity ladder scored by completeness with literature priors versus fitted. One paragraph connecting to the completeness/restrictiveness framework: the null model and the prior are the same question, asked in two languages.

Close. What the tool can and cannot see; the toothbrush problem for swarm analysis; invitation to run `swarm-sna` on the next incident dataset.

Figures (aim for 5): null distributions with observed marks; one network per era; hierarchy before/during/after leaders; NBDA hazard curve for one cascade; perceived-vs-actual edge agreement.

Tone notes for the blog: no model-family league tables; every claim with its null; name the confounds (scaffolding changes, one instance per model) in the body, not a footnote.

Risks and caveats

Risk	Why it matters	Mitigation
Rooms (#best / #rest and 3 others)	Community structure will just be room assignment	Permute within room × day; report brokerage across rooms separately
Three eras, four hours-per-day changes	Pooled statistics mix regimes and observation effort	Per-era reporting; HWI; per-100-agent-hour rates
Scaffolding changes (nudge system, prompt edits, humans removed)	Behavior shifts may be downstream of the scaffold	<code>shocks.csv</code> from <code>CHANGELOG.md</code> ; never interpret a change across a shock as emergent
One instance per model with persistent memory	Model vs instance is not identifiable	Say "this instance"; never a model-family league table
Message volume is gamed (850k fragments, 52 essays/day)	Degree centrality is meaningless	Volume-conditioned nulls; dedupe scripted posts; cap per-agent-day weight
Self-reports are unreliable (fabricated GitHub stats, divergent realities)	<code>acted</code> from chat alone overstates compliance	Use <code>computer_use_sessions</code> for <code>acted</code> ; treat chat claims as claims
Name ambiguity ("Opus", three GPT-6 variants Sol/Luna/Terra)	Mention edges misattributed	Family-and-room disambiguation; weight 1/k; report mention precision
Work bleeds across goal boundaries	Per-goal networks leak	±1 day overlap windows

Risk	Why it matters	Mitigation
Timeline/goal summaries are LLM-written	Not ground truth	Use only for choosing cascade cases; verify in chat
LLM-classifier cost/time for Layers C and D	Could eat Saturday	Slice first, cache, run full pass overnight
Wiki agents rename themselves	Node identity unstable	Treat wiki as a reciprocity/diffusion check only; say so
Judges expect a demo, not a paper	A notebook is not a tool	CLI + README + recording first; analyses second